

TASK 3 — Pharmacovigilance & ADR Analysis

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Pharmacovigilance Report on Ibuprofen

Introduction

Ibuprofen is a non-steroidal anti-inflammatory drug (NSAID) widely used for pain relief, fever reduction, and inflammation management. Although highly effective, long-term or excessive use may result in adverse drug reactions (ADRs), making pharmacovigilance monitoring essential.

Mechanism of Action

Ibuprofen inhibits cyclooxygenase enzymes:

- COX-1
- COX-2

This inhibition reduces prostaglandin synthesis, thereby decreasing:

- Pain
- Inflammation
- Fever

However, suppression of prostaglandins may also affect gastric mucosa, renal blood flow, and platelet function.

Common Therapeutic Uses

Therapeutic Use	Application
Analgesic	Headache, muscle pain
Antipyretic	Fever reduction

Therapeutic Use	Application
Anti-inflammatory	Arthritis and swelling
Post-operative care	Pain management

Reported Adverse Drug Reactions (ADRs)

Gastrointestinal ADRs

- Gastric irritation
- Peptic ulcer formation
- Gastrointestinal bleeding
- Nausea and vomiting

Renal ADRs

- Reduced renal perfusion
- Acute kidney injury
- Fluid retention

Cardiovascular ADRs

- Elevated blood pressure
- Increased cardiovascular risk
- Edema

Allergic Reactions

- Rash
- Bronchospasm
- Hypersensitivity reactions

Neurological Effects

- Dizziness
 - Headache
 - Drowsiness
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ADR Sources and Pharmacovigilance Databases

ADR information has been collected from:

- WHO Vigibase
- DrugBank
- FDA Adverse Event Reporting System
- Published clinical studies

Pharmacovigilance systems help identify rare, severe, or long-term drug safety concerns.

Risk Factors

Certain populations are more vulnerable to ADRs:

Risk Group	Increased Concern
Elderly patients	GI bleeding
Renal disease patients	Kidney toxicity
Cardiovascular patients	Blood pressure effects
Asthma patients	Bronchospasm risk

Measures to Reduce Risks

Clinical Measures

- Use lowest effective dose
- Limit duration of therapy
- Avoid unnecessary long-term use
- Monitor renal function
- Assess GI risk factors

Patient Education

Patients should:

- Avoid overdose
- Take medication after meals
- Report unusual symptoms
- Avoid combining multiple NSAIDs

Pharmaceutical Strategies

- Enteric coated formulations
- Combination therapy with gastroprotective agents
- Improved labeling and warnings

Importance of Pharmacovigilance

Pharmacovigilance plays a major role in:

- Detecting rare ADRs
- Improving patient safety
- Updating prescribing guidelines
- Supporting regulatory decisions
- Enhancing public health outcomes

Modern pharmacovigilance increasingly uses:

- AI-based signal detection
- Real-world evidence analysis
- Big data monitoring
- Electronic healthcare records

Conclusion

Ibuprofen remains an effective and widely used NSAID; however, continuous pharmacovigilance monitoring is essential to ensure safe therapeutic use.

Through proper risk assessment, patient education, and regulatory surveillance, healthcare professio